Hints: Probability – 1. Expectation, indicators and linearity

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. A run of $k+1$ equal symbols, heads or tails, is the same as k consecutive "repeats" in the sequence $d_i = [x_i = x_{i+1}]$, $i = 1, \dots, n-1$.

Hint 2. The map $x \mapsto d$ is 2-to-1 onto $\{0,1\}^{n-1}$, so d is a sequence of $n-1$ fair tosses and $\Pr(B)$ is the probability of k consecutive heads in $n-1$ tosses.

Hint 3. This is at most $\Pr(A)$ (look at the first $n-1$ of the n tosses), and strictly less, because $T^{n-k}H^k$ belongs to A while its first $n-1$ tosses do not. Answer: A is more probable.

Problem 2

Hint 1. Set up states by the lead and the server, with $q = 1-p$: (24:22, leader serves), (24:23, trailer serves), tie with someone serving, (lead 1, leader serves). Observe that from a tie the next rally always produces "lead 1, leader serves", whoever wins.

Hint 2. For the tie D and lead-one L: $E_D = 1 + E_L$, $E_L = 1 + qE_D$, so $E_D = 2/p$. Then $E_{24:23} = 1 + pE_D = 3$ and $E_{24:22} = 1 + qE_{24:23}$.

Hint 3. Answer: $4 - 3p$ for $0 < p \leqslant 1$. (For $p = 0$ the set ends after exactly 2 rallies.)

Problem 3

Hint 1. Only the current interval of candidates matters, and the hidden number is uniform on it. Let $V(n)$ be n times the winning probability of the player to move when n candidates remain; guessing a number that splits the rest into a and $b = n-1-a$ gives $V(n) = \max_{a+b=n-1}\bigl(n - V(a) - V(b)\bigr)$ with $V(0) = 0$.

Hint 2. Compute small cases and prove by induction that $V(n) = \lceil n/2 \rceil$: $V(a) + V(b) \geqslant \lceil (n-1)/2 \rceil = \lfloor n/2 \rfloor$, with equality for $a = 0$ (guess an endpoint). Wasting a move on an already excluded number gives only $n - V(n) \leqslant V(n)$.

Hint 3. Answer: $\frac{1}{2}$ for even N and $\frac{N+1}{2N}$ for odd N.

Problem 4

Hint 1. Lower bound for Vasya: let η be an independent copy of ξ. Then $\Pr(\eta \geqslant \xi) + \Pr(\xi \geqslant \eta) \geqslant 1$, and both are equal.

Hint 2. Upper bound for Petya: take ξ uniform on $[0,2]$. Then $\Pr(\eta \geqslant \xi) = \mathrm{E}F(\eta)$ with $F(t) = \Pr(\xi \leqslant t) = \min(t/2, 1) \leqslant t/2$, so it is at most $\mathrm{E}\eta/2$. Answer: $p = \frac{1}{2}$.

Problem 5

Hint 1. Track the number k of white balls. A step changes k only if a mixed pair is drawn (probability $r_k = k(n-k)/\binom{n}{2}$), and then $k \to k \pm 1$ with probability $\frac{1}{2}$ each: a simple symmetric random walk on $\{0, \dots, n\}$ from 1, slowed down at each state.

Hint 2. The expected number of visits of the walk to j before absorption, starting from 1, is $G(1,j) = \frac{2(n-j)}{n}$; each visit lasts on average $1/r_j$ steps.

Hint 3. $\mathrm{E}T = \sum_{j=1}^{n-1} \frac{2(n-j)}{n} \cdot \frac{n(n-1)}{2j(n-j)} = (n-1)H_{n-1}$.

Problem 6

Hint 1. By linearity, $\mathrm{E}S = \sum_i 2^{-i}\,\mathrm{E}\big[X_i \mathbf{1}_{\{k \geqslant i\}}\big]$, and $k \geqslant i$ iff $X_1 \geqslant X_2 \geqslant \ldots \geqslant X_i$.

Hint 2. Conditioning on $X_i = x$: $\Pr(X_1 \geqslant \ldots \geqslant X_{i-1} \geqslant x) = \frac{(1-x)^{i-1}}{(i-1)!}$, so the i-th term equals $\frac{2^{-i}}{(i-1)!}\int_0^1 x(1-x)^{i-1}dx = \frac{2^{-i}}{(i+1)!}$.

Hint 3. $\mathrm{E}S = \sum_{i \geqslant 1} \frac{2^{-i}}{(i+1)!} = 2\big(e^{1/2} - 1 - \frac{1}{2}\big) = 2e^{1/2} - 3$.

Problem 7

Hint 1. Wald: $X_1 + \cdots + X_N = \sum_i X_i \mathbf{1}_{\{N \geqslant i\}}$, and $\{N \geqslant i\} = \{N \leqslant i-1\}^c$ is determined by $X_1, \ldots, X_{i-1}$, hence independent of X_i. So the expectation is $\sum_i \mathrm{E}X_1 \Pr(N \geqslant i) = \mathrm{E}X_1\,\mathrm{E}N$ (justify the exchange by doing the same with $|X_i|$).

Hint 2. Pattern of length m: before each toss a new gambler arrives with \$1 and bets it on the pattern, doubling at each correct toss and leaving when a toss is wrong. The game is fair, and $\mathrm{E}N < \infty$ (disjoint blocks of m tosses each give the pattern with probability 2^{-m}), so the casino's expected net gain at time N is 0.

Hint 3. At time N the casino has received N dollars and pays 2^j to the gambler who started j tosses earlier exactly when the prefix of length j of the pattern equals its suffix of length j. So $\mathrm{E}N = \sum_{j:\,\mathrm{prefix}_j = \mathrm{suffix}_j} 2^j$: $2 + 4 = 6$ for HH and 4 for HT.

Problem 8

Hint 1. Follow $1 \to \sigma(1) \to \sigma^2(1) \to \ldots$: at step ℓ the cycle closes with probability $\frac{1}{n-\ell+1}$ given that it is still open, so $\Pr(L = \ell) = \frac{1}{n}$ for each ℓ and $\mathrm{E}L = \frac{n+1}{2}$.

Hint 2. The number of cycles is $\sum_i 1/L_i$, with expectation $n\,\mathrm{E}(1/L_1) = H_n$. Asymptotics: $\mathrm{E}L \sim n/2$ and $H_n = \ln n + \gamma + o(1)$.

Problem 9

Hint 1. Place the three circles and take, for each pair, the common external tangent on the side away from the third circle. A triangle exists iff these three lines enclose the circles, i.e. all three interior angles come out positive. The angle at the vertex near the largest circle (radius t) is the critical one.

Hint 2. With β the angle of the triangle of centres at the centre of the t-circle, the angle at that vertex is $\beta - \theta_r - \theta_s$ with $\sin\theta_r = \frac{t-r}{t+r}$. Using $\sin^2\frac{\beta}{2} = \frac{rs}{(t+r)(t+s)}$ and $\theta_r = \frac{\pi}{2} - 2\arctan\sqrt{r/t}$, positivity becomes $t^2 < 4rs$.

Hint 3. So the condition is $\max^2 < 4 \cdot (\text{product of the other two})$; only the largest part matters. Over the simplex $r + s + t = 1$ with t largest, the admissible r form an interval of length $3t - 1$ for $\frac{1}{3} \leqslant t \leqslant \frac{4}{9}$ and $\sqrt{1-2t}$ for $\frac{4}{9} \leqslant t \leqslant \frac{1}{2}$. The probability is $6\big(\int_{1/3}^{4/9}(3t-1)\,dt + \int_{4/9}^{1/2}\sqrt{1-2t}\,dt\big) = 6\big(\frac{1}{54} + \frac{1}{81}\big) = \frac{5}{27}$.

Problem 10

Note. Read literally, the tail bound can fail when $n\ln n + cn$ is not an integer (e.g. $n = 2$, $c = 0.1$: the time is always $\geqslant 2 > 2\ln 2 + 0.2$). The standard statement, followed here, is that after

$m \geqslant n \ln n + cn$ draws (an integer number m) the probability that some coupon is still missing is at most e^{-c}.

Hint 1. When $i-1$ coupons have been collected, the waiting time for a new one is geometric with success probability $\frac{n-i+1}{n}$, so its mean is $\frac{n}{n-i+1}$. Summing gives nH_n.

Hint 2. Tail: union bound over coupons. $\Pr(\text{some coupon missing after } m \text{ draws}) \leqslant n(1-\frac{1}{n})^m \leqslant ne^{-m/n} \leqslant e^{-c}$ for $m \geqslant n \ln n + cn$.